GTU Applied Mathematics (DI02000011)

Unit 5: Statistics

Topic 5.1: Mean for Ungrouped and Grouped Data

Definition: The arithmetic mean (average) is the central value of a data set. It is found by adding all data values together and dividing the sum by the total number of items. Ungrouped data is a raw list of numbers, while grouped data is organized into frequency tables.

3 Key Features:

- **Central Representative:** It acts as a balance point where the sum of distances to all data points below it equals the sum of distances to all data points above it.
- **Formula Shift:** For raw ungrouped lists, the formula is $mean = \sum x / n$. For grouped frequencies, it shifts to $mean = \sum fx / \sum f$.
- **Sensitivity:** The mean is highly sensitive to extreme high or low values (outliers), which can pull the average away from the true center.

3 Major Uses:

- **Quality Assurance:** Calculating the average lifespan of a batch of manufactured light bulbs or car components.
- **Performance Testing:** Finding the average structural strength of a concrete blend after testing several sample cubes.
- **Resource Planning:** Evaluating average daily power or water consumption to balance utility grids.

Real-Life Application:

Think of your **semester exam marks**. If you list your individual percentages for 5 subjects, that is ungrouped data. Adding them up and dividing by 5 gives you your average score, representing your overall performance.

Topic 5.2: Mean Deviation and Standard Deviation

Definition: These metrics measure dispersion, which tells you how spread out or scattered your data points are around the mean average. Mean Deviation calculates the average absolute distance from the mean, while Standard Deviation is a more powerful tool that squares those distances to highlight larger variations.

3 Key Features:

- **Consistency Gauge:** A low standard deviation means data points are tightly packed around the average (highly consistent); a high value means the data is widely spread out (unreliable).
- **The Variance Connection:** Squaring the standard deviation gives you the 'variance', another core value used in statistical engineering.

- **Positive Focus:** Mean deviation ignores negative signs by using absolute brackets, while standard deviation clears negative values by squaring them.

3 Major Uses:

- **Manufacturing Tolerances:** Monitoring automated cutting machines to ensure mechanical parts do not vary too much from the required blueprint thickness.
- **Risk Assessment:** Evaluating the reliability of electronic sensors by checking how much their voltage readings fluctuate in changing temperatures.
- **Traffic Engineering:** Analyzing speed variations on a highway to determine if erratic driving patterns are creating safety hazards.

Real-Life Application:

Imagine two automated **beverage bottling lines**. Both lines fill bottles with an average (mean) of 500 ml. However, Line A has a very low standard deviation (filling between 499 ml and 501 ml), while Line B has a high standard deviation (filling between 490 ml and 510 ml). Line A is the superior, more precise machine.